

SAMARA PIRES

Applied AI | Machine Learning | Data Science | Data Analytics

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Professional Experience

AltaML - Associate Business Solution Consultant

May 2026 - Aug 2026

- Bridged **business and technical teams** in a hybrid Product Management–AI Architecture capacity, translating stakeholder needs into AI solution requirements, workflows, and product recommendations.
- Contributed to an **LLM-powered Research Assistant for the Government of Alberta**, gathering requirements, evaluating solution performance, and translating client feedback into product improvements.
- Collaborated with AI engineers to design, test, and refine **agentic AI workflows** using **LLMs, retrieval-based approaches, prompt engineering, and system evaluation**.
- Supported the **end-to-end AI product lifecycle**, spanning discovery and requirements gathering through solution design, UAT, deployment, and iterative improvement.

Amii (Alberta Machine Intelligence Institute) – AICAP : Level Up WILO Participant

Sep 2025 - Present

- Selected for Amii's **Level Up Work-Integrated Learning Opportunity (WILO)**, working alongside an Amii Machine Learning Scientist to analyze real-world AI challenges faced by startups and evaluate practical solution approaches.
- Conducted technical research across **LLM agents, human-in-the-loop systems, human feedback, prompt/context engineering, and agentic workflows**, translating findings into detailed technical reports and recommendations for participating startups.
- Participated in Amii's **AI Career Accelerator Program (AICAP)** through technical AI/ML coursework, industry-led professional development, and engagement with Alberta's AI ecosystem.

Data For Good Edmonton - Data Analyst

Sep 2025 – May 2026

- Built a **Python/NumPy cost-benefit modelling pipeline** to estimate the **10-year health and economic impact** of Edmonton's protected cycling infrastructure using cyclist-count data, epidemiological risk reductions, and healthcare cost parameters.
- Conducted **multi-scenario CVD and diabetes cost-savings analysis** and developed **Matplotlib visualizations** to compare projected healthcare benefits against infrastructure capital and operating costs.

LTIMindtree - Quality Analyst | Client: Johnson Controls

Jun 2021 - Oct 2024

- Developed **automated testing frameworks using Python, Selenium, and BrowserStack**, reducing **manual testing effort by 81%** and improving testing efficiency across enterprise applications.
- Led **cross-functional delivery** for Johnson Controls' **Customer Self-Service Portal (UK & Ireland)**, coordinating QA, development, and business stakeholders to achieve a **5/5 client rating** for timely, high-quality delivery.
- Consistently delivered across a multi-year client engagement, earning **Outstanding Performer recognition**, a **4/4 performance rating**, and **100% client satisfaction**.
- Facilitated **company-wide town halls for 600+ employees**, strengthening cross-team collaboration, knowledge sharing, and alignment across technical and non-technical groups.

Expenzing - Data Analyst Intern

Dec 2019 - Jan 2020

- Built **interactive Power BI dashboards** for expense management, translating transactional data into **actionable financial insights** that contributed to **23% cost savings**.
- Performed **data cleaning, validation, and workflow automation** on payment and expense datasets, improving **process efficiency by 32%** and reducing manual intervention.
- Presented **financial trends, spending patterns, and anomalies** through data visualizations and management-facing insights to support data-driven expense decisions.

Education

University of Alberta - Master of Science in Modelling, Data & Predictions | 2025 – Dec 2026 | GPA: 3.7 / 4.0

Relevant Coursework: Mathematical Modelling for Finance & Government; Statistical Analysis for Research; Mathematical Modelling for Infectious Diseases

University of Mumbai, Don Bosco Institute of Technology - Bachelor of Engineering, Computer Science | 2017 - 2021

Coursework: Machine Learning; Statistical Analysis; Data Visualization; Python Programming

Award: Runner – Up, Best Final Year Thesis

Technical Skills

Programming & Data: Python, SQL, R, MATLAB, Pandas, NumPy

Machine Learning & Statistics: Scikit-learn, Predictive Modeling, Statistical Modeling, Bayesian Statistics, PyMC, Model Evaluation

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Semantic Search, FAISS

Analytics & Visualization: Exploratory Data Analysis (EDA), Data Cleaning, Excel, Power BI, Tableau, Matplotlib

Tools & Platforms: Git, GitHub, Jupyter, Streamlit, Google BigQuery

Technical Projects

2026 Data Masters Challenge - 2nd Place: [Link](#)

Python | Pandas | NumPy | Scikit-learn | Machine Learning

- Built an end-to-end **machine learning pipeline** using Python, Pandas, NumPy, and Scikit-learn, encompassing data cleaning, EDA, feature preparation, predictive modeling, and model evaluation.
- Evaluated and visualized model performance to generate actionable insights, contributing to a **2nd-place finish in the 2026 Data Masters Challenge**.

Budget Bites - RAG-Powered AI Recipe Planner: [Link](#)

Python | LLMs | RAG | FAISS | Streamlit

- Developed a **Retrieval-Augmented Generation (RAG)** recipe recommendation system using **LLMs, FAISS, and semantic search** to retrieve relevant recipe context and generate personalized recommendations.
- Built an interactive **Streamlit application** with retrieval and prompt-engineering workflows to generate recipe suggestions based on user preferences and available ingredients.

BNPL Risk Prediction - Bayesian Machine Learning: [Link](#)

Python | PyMC | Bayesian Statistics | Predictive Modeling

- Developed a **Bayesian machine learning model** for Buy Now, Pay Later (BNPL) risk prediction using **Python and PyMC**, applying probabilistic modeling to estimate customer risk.
- Evaluated **predictive performance and uncertainty** to generate interpretable risk estimates for financial risk assessment.

Research & Publications

Cyber Threat Analysis Using Pearson & Spearman Correlation via EDA - ICSCCC 2023, IEEE

Statistical modeling and EDA on **UNSW-NB15** using Pearson/Spearman correlation, chi-square and t-tests for cyberattack analysis.

A Real-Time Position Monitoring System for Fall Detection & Analysis - ICNTE 2021, IEEE

Developed **OpenPose/SSD-MobileNet** fall-detection models achieving **93% classification accuracy**.

Fall Stop: Real-Time Fall Detection & Vital Monitoring - INDISCON 2021, IEEE

Developed an **SVM-based IoT fall-detection system** integrating real-time sensor data and automated alerts.

Leadership & Community

DBIT SevaTech - Co-Founder: Led a student technology initiative delivering digital solutions for NGOs and creating **40+ student internship opportunities**.

SIGMAS - Outreach Coordinator: Coordinated mathematical outreach across Edmonton schools, including a **900+ student Math Fair/Unfair**.